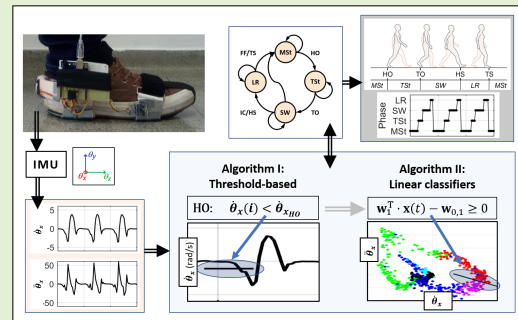


Identification of Gait Events in Healthy and Parkinson's Disease Subjects Using Inertial Sensors: A Supervised Learning Approach

Juan C. Pérez-Ibarra¹, Adriano A. G. Siqueira², *Member, IEEE*, and Hermano I. Krebs, *Fellow, IEEE*

Abstract—Automatic detection of gait phases through supervised learning is a feasible approach that takes advantage of the consistency of the gait cycle among healthy subjects. However, gait patterns among subjects with impairments are not as consistent and most of the existing algorithms have limited performance detecting phases during impaired gait. In this paper, we proposed one algorithm that used linear classifiers to detect in real-time the transition between consecutive gait phases. Our approach is a generalization of the rule- and threshold-based algorithms for event detection. Linear classifiers are parametric models that require appropriate values in order to perform correct classification of the gait phases. We introduced a modified Support Vector Machine (SVM) to compute such sub-optimal combinations of those values, and a further optimization with a hybrid meta-heuristic approach that integrates a Genetic and a Simulated Annealing Algorithm. We tested our approach on data collected by a single-IMU foot-mounted wearable device during overground and treadmill walking for two groups: one with healthy and one with Parkinson's Disease subjects. The F_1 -scores were 0.987 and 0.953 for the two groups, which were comparable with our previously developed threshold-based method, which obtained 0.988 and 0.974, respectively. Our proposed approach achieved similar performance as the threshold-based scheme, with the advantage of not relying on any prior knowledge of specific features for any particular inertial signal.

Index Terms—Gait analysis, gait phases, wearable sensors, machine learning, meta-heuristics.



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I. INTRODUCTION

GAIT analysis, defined as the systematic study of human locomotion, can improve the process of gait rehabilitation [1]. Proper identification of gait events permits control of assistive or rehabilitation robotic [2] and electrical stimulation devices [3]. Ultimately we want to determine in real-time the occurrence of the gait events using the smallest amount of data and computational cost. Wearable sensors are feasible alternatives that have been increasingly used for gait analysis, and especially for detection of gait phases [4]. Force-Sensitive Resistors (FSR) are a wearable, very efficient alternative that provides reliable information about the gait phases [5]. However, available pressure sensors have low mechanical robustness and limitation in terms of the total number of strides. Inertial sensors, like gyroscopes and accelerometers, are robust, cheap, small, easy to place, require low power, and provide reliable info about human gait that can be used for the identification of gait patterns and detection of events [6]. Automatic detection of gait phases has been previously implemented following two main approaches: 1) use of predefined rules [7], [8], and 2) use of machine learning algorithms [9], [10]. Both approaches are based on the search for specific values in the gait data attempting to characterize each phase or transition. Kinematics during healthy gait has been extensively characterized and given thresholds can be used to detect events [11]. A common approach that takes advantage of the known features of gait

signals is the dual-minima approach. It detects gait events using the min-max-min sequence observed at sagittal angular velocity of the foot [12]. However, gait patterns in subjects with gait disorders are not as consistent [13], [14], and their high variability limits the use of algorithms developed to detect events for healthy subjects' gait [15]. In consequence, there is a lack of suitable algorithms for real-time identification of gait phases in subjects with gait disorders using data from foot-mounted inertial sensors.

Parkinson's disease (PD) is one of the most common neuro-degenerative diseases, and it is a main source of gait disorder in the older population. PD produces continuous and episodic gait disturbances that range from reduction of walking speed to risk of falls [14]. Detection of gait events is highly relevant in the gait analysis of PD subjects, since it enables the detection of freezing-of-gait episodes [16] as well as the evaluation of the stride-to-stride variability [17]. Despite the great interest in this topic, demonstrated by a large number of papers that report gait event detection algorithms for subjects with PD [18]–[27], most of those correspond to offline implementations that are not suitable for real-time application. In [28] and [29], authors used the dual-minima approach for identification of two gait events in PD subjects. However, they proposed “soft” real-time implementations that require a fixed delay of half a stride. Significant progress was made in [30], where real-time detection of three gait events was performed using an algorithm based on a moving Discrete Wavelet Transform, which is trained offline using the algorithm in [18]. Recently, in [31], we proposed an algorithm for detection of two gait events that integrated the observed advantages of the algorithms developed in [32] and [33]. Here, we aim to expand those rules to the detection of four gait events.

Our research aims to develop accurate and efficient methods for real-time detection of gait events in subjects with gait disorders. In this paper, we propose and evaluate two new algorithms for the identification of four gait events in healthy and PD subjects, using data collected with a foot-mounted inertial measurement unit (IMU) [31]. First, we introduce a rule-based algorithm that summarizes most of the approaches used in the literature for gait segmentation. Then, we propose a generalization of the threshold-based rules by using linear classifiers to determine gait phase transitions and a parameter learning and optimization procedure to determine the most adequate values for the parameters of both the thresholds and the linear classifiers for each subject. The proposed algorithm offers the benefit of not having any previous knowledge of the characteristics or patterns of the evaluated gait signals.

II. GAIT PHASES

Gait is a cyclic pattern segmented in a diverse number of phases. A simple approach is to divide gait into two phases: Stance and Swing. One can divide Stance and Swing in additional sub-phases reaching up to the eight phases of the gait model proposed by the Rancho Los Amigos Gait Committee: initial contact, loading response, mid-stance, terminal stance, pre-swing, initial swing, mid-swing, and terminal swing [34].

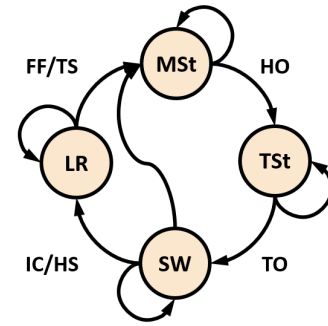


Fig. 1. Gait modeled as a 4-state FSM where each state corresponds to one of the following phases: Mid-Stance (MSt), Terminal Stance (TSt), Swing (SW), and Loading Response (LR). Transitions between phases correspond to the events: IC/HS, FF/TS, HO, and TO.

Gait events define the transitions among phases. For instance, the initial and final contacts between foot and ground delimit stance and swing. During normal gait, heel is the first part of the foot to contact the floor. Therefore, Initial Contact (IC) corresponds to the Heel-Strike (HS) event. Similarly, End Contact (EC) corresponds to the Toe-Off (TO) event, since the toe is the last part of the foot to lose contact with the floor. Two more events during normal gait are Heel-Off (HO), when the heel rises and the limb moves forward, and Toe-Strike (TS), when the toe contacts the ground. It is important to consider that these descriptions often fail when applied to pathological gait. For example, when the heel and toe hit the ground at the same time without a precedent HS.

III. ALGORITHMS FOR ONLINE IDENTIFICATION OF GAIT PHASES AND EVENTS

In this section, we propose two online causal algorithms for gait event detection and phase segmentation. We model gait as a finite state machine (FSM) where each state corresponds to a gait phase and transitions between them correspond to gait events (see Fig. 1).

A. Algorithm 1: Heuristic Threshold-Based Event Detection

Consider an inertial sensor measuring 3D angular velocities ($\dot{\theta}_x, \dot{\theta}_y, \dot{\theta}_z$) and linear accelerations of the foot (a_x, a_y, a_z) during gait. To determine transitions among states, we propose an algorithm that tests at each time sample one of the following rules depending on the current state, z :

- During Mid-Stance, $z_{(i)} = 1$, the algorithm moves to the terminal stance phase, $z_{(i+1)} = 2$, if the sagittal angular velocity, $\dot{\theta}_x$, and acceleration, $\ddot{\theta}_x$, are below the negative thresholds, $\dot{\theta}_{xHO}$ and $\ddot{\theta}_{xHO}$, respectively. The rationality behind this rule is that during a small time after HO, the forefoot continues on the ground causing plantar flexion of the foot.
- During Terminal-Stance, $z_{(i)} = 2$, the algorithm moves to the swing phase, $z_{(i+1)} = 3$, if $\dot{\theta}_x$ is above a threshold $\dot{\theta}_{xTO}$. The foot is in dorsiflexion when it is leaving the ground and following the leg during swing.
- During Swing, $z_{(i)} = 3$, the algorithm moves to the loading response phase, $z_{(i+1)} = 4$, if $\dot{\theta}_x$ and the horizontal linear acceleration a_z are respectively below the thresholds $\dot{\theta}_{xHS}$ and a_{zHS} . End of the swing phase combines two factors

Algorithm 1 Heuristic Threshold-Based Method

Input: $\dot{\theta}_x, \ddot{\theta}_x, a_z, \dot{\theta}_r$
1: Initialize $z(1), \dot{\theta}_{xHO}, \ddot{\theta}_{xHO}, \dot{\theta}_{xTO}, a_{zHS}, \dot{\theta}_{xHS}, \dot{\theta}_{rTS}, \delta_{TS}^{lag}$
2: **for** $i = 2$ **to** n **do**
3: $z(i) = z(i-1)$
4: **switch** $z(i)$ **do**
5: **case** 1 - (*Mid-stance - Detection of HO*) :
6: **if** $\dot{\theta}_x(i) < \dot{\theta}_{xHO}$ **and** $\ddot{\theta}_x(i) < \ddot{\theta}_{xHO}$ **then**
7: $z(i+1) = 2, HO(i) = 1$
8: **end if**
9: **case** 2 - (*Terminal Stance - Detection of TO*) :
10: **if** $\dot{\theta}_x(i) > \dot{\theta}_{xTO}$ **then**
11: $z(i+1) = 3, TO(i) = 1$
12: **end if**
13: **case** 3 - (*Swing - Detection of MSw and HS*) :
14: **if** $\ddot{\theta}_x(i) = 0$ **and** $\dot{\theta}_x(i) > 0$ (*i.e., max. in $\dot{\theta}_x$*) **then**
15: $MSw = 1$
16: **end if**
17: **if** $a_z(i) < a_{zHS}$ **and** $\dot{\theta}_x(i) < \dot{\theta}_{xHS}$ **and** $MSw = 1$ **then**
18: $z(i+1) = 4, HS(i) = 1$
19: **end if**
20: **case** 3 or 4 - (*Loading Response - Det. of TS*) :
21: **if** $\dot{\theta}_r(j) < \dot{\theta}_{rTS} \forall j \in [i - \delta_{TS}^{lag}; i]$ **then**
22: $z(i+1) = 1, TS(i) = 1$
23: **end if**
24: **end switch**
25: **end for**
Output: HO, TO, HS, TS, z

that we aim to detect with those rules: the foot slows down in its forward advance and the foot increases its plantar flexion angle after the heel hits the ground. We included an additional rule to prevent the wrong detection of HS, limiting it to be detected only after an additional event known as mid-swing, MSw, corresponding to a positive peak of $\dot{\theta}_x$ during swing.

- During Loading Response, $z(i) = 4$, the system looks for the gait period in which the foot is flat over the ground. To do that, we look at the resultant angular velocity signal computed as $\dot{\theta}_r = \sqrt{\dot{\theta}_x^2 + \dot{\theta}_y^2 + \dot{\theta}_z^2}$. The beginning of the mid-stance phase is detected when this signal is below a given threshold $\dot{\theta}_{rTS}$ during a time longer than δ_{TS}^{lag} .

As mentioned earlier, during pathological gait, the heel and toe may make initial contact at the same time. To detect those cases, the algorithm also performs detection of the mid-stance phase during swing. The pseudo-code of Algorithm 1 implements the proposed algorithm and its rules.

B. Algorithm II: Linear Classifiers for Gait Event Detection

Let us express each of the previously defined threshold-based rules in the form:

$$EV(i) = \begin{cases} 1, & \text{if } w_z x(i) > b_z \\ 0, & \text{otherwise,} \end{cases}$$

Algorithm 2 Method of the Linear Classifiers

Input: $\mathbf{x}$
1: Initialize $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, z(1), \delta_1, \delta_2$
2: **for** $i = 2$ **to** n **do**
3: $z(i) = z(i-1)$
4: **if** $\mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z \geq 0, \forall i \in [i - \delta_1, i]$ **then**
5: $EV(i, z) = 1, z(i+1) = z(i) + 1^*$
6: **end if**
7: **end for**
Output: EV, z
* Next value in the cyclic sequence $z = \{1, 2, 3, 4, 1, 2, \dots\}$.

where $EV(i) = [HO, TO, HS, TS]^T$ is a 1-of-k vector, with all zeros except for one element that indicates an state transition away from state $z \in \{1, 2, 3, 4\}$, $x(i)$ is a given feature, $w_z \in \{+1, -1\}$ is a weight that determines the order of the inequality for each state z , and b_z is a threshold used for the feature x at each phase z .

Having multiple features, we can reformulate the rule for each phase as a linear classifier given by:

$$y(\mathbf{x}_i) = \begin{cases} +1 & \text{if } \mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z \geq 0 \\ -1 & \text{if } \mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z < 0, \end{cases} \quad (1)$$

where $y = +1$ indicates a state transition, $\mathbf{w}_z$ and $\mathbf{b}_z$ are vectors of weight parameters (which generalize thresholds) for the classifier of the state $z(i)$, and $\mathbf{x}_i$ is the feature vector at time i . If $\mathbf{x}$ is univariate, the linear classifier is equivalent to a threshold-based rule. When $\mathbf{x}$ is multivariate, the linear classifier becomes a linear combination of the features. To avoid undesired effect from noise and artifacts, we define a lag time δ so that we only allow transitions after the lag.

Finally, we can express the linear classifier by:

$$y(\mathbf{x}_i) \mapsto \text{sgn}(\mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z) \quad \forall i \in [i - \delta, i]. \quad (2)$$

This generalization allows the use of any signal as input of the classifier and the use of optimization/learning procedures to find (sub) optimal parameters, reducing the need for domain knowledge. Our proposal is to switch between different linear classifiers, one for each phase, so that when the FSM moves to a new state, the classifier changes (see pseudo-code of Algorithm 2 below).

C. Parameter Learning Using Support Vector Machines

To learn adequate parameters for the classifiers, we propose a modified implementation of the soft-margin Support Vector Machine (SVM) algorithm. Consider data from two consecutive states z_1 and z_2 , then set the label $t_i = -1$ if $z = z_1$ and $t_i = +1$ if $z = z_2$. We use SVM to maximize the margin between the decision function, $\mathbf{w}_z^T \mathbf{x} - \mathbf{b} = 0$, and the data samples of the training set previously labeled. Since vector $\mathbf{w}_z$ is normal to the decision function, maximization of the margin is equivalent to the minimization problem:

$$\begin{aligned} \arg \min_{\mathbf{w}_z} \quad & \frac{1}{2} \|\mathbf{w}_z\|^2 + C \sum_{i \in \mathcal{Z}} \xi_i, \\ \text{s.t.} \quad & t_i \cdot (\mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z) \geq 1 - \xi_i \quad \text{and} \quad \xi_i \geq 0, \text{ for } i \in \mathcal{Z} \end{aligned} \quad (3)$$

where $\mathcal{Z}$ denotes the set of indices with $z = \{z_1, z_2\}$; $\xi_i \geq 0$, are slack variables for each point i added to penalize the distance of the point to the boundary; $C > 0$ is a scalar that controls the trade-off between the margin and the penalties.

Using dual Lagrangian optimization, the solution of (3) is given by:

$$\mathbf{w}_z = \sum_{i \in \mathcal{Z}} a_i t_i \mathbf{x}_i, \quad \text{and} \quad \sum_{i \in \mathcal{Z}} a_i t_i = 0 \quad (4)$$

where $a_i \geq 0$ are Lagrange multipliers for each point which are computed by solving the quadratic optimization problem:

$$\max_a \sum_{i,j \in \mathcal{Z}} a_i a_j t_i t_j \mathbf{x}_i^T \mathbf{x}_j \quad (5)$$

satisfying the Karush-Kuhn-Tucker (KKT) conditions:

$$\begin{aligned} a_i &\geq 0; \quad t_i \cdot (\mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z) - 1 \geq 0; \\ a_i \cdot \{t_i \cdot (\mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z) - 1\} &= 0. \end{aligned}$$

so that for every data point, $a_i = 0$ or $t_i \cdot (\mathbf{w}_z^T \mathbf{x}_i - \mathbf{b}_z) = 1$. Points that satisfy the latter condition lie on the maximum margin boundary and are called Support Vectors.

New points are classified using the following model expressed in terms of the parameters a_n ,

$$y(\mathbf{x}_i) \mapsto \text{sgn} \left(b_z + \sum_{j \in \mathcal{S}} \underbrace{a_j y_j \mathbf{x}_j^T \mathbf{x}_i}_{\mathbf{w}_z^T \mathbf{x}_i} \right), \quad (6)$$

$$b_z = \frac{1}{N_S} \sum_{i \in \mathcal{S}} \left(y_i - \sum_{j \in \mathcal{S}} a_j y_j \mathbf{x}_j^T \mathbf{x}_i \right), \quad (7)$$

where $\mathcal{S}$ denotes the set of indices for the support vectors, and N_S is the number of support vectors.

D. Parameter Optimization Using Metaheuristics

Consider that both the previous algorithms use parametric models for gait event detection: either the set of thresholds of the Algorithm I, or the weights $\mathbf{w}_z$ and $\mathbf{b}_z$ of the Algorithm II. Our hypothesis is that for these models, given a predefined model structure (number of phases, allowed transitions and feature vector), there is a set of parameters that allows maximum accuracy in the gait event detection.

We propose a global optimization of the parameter set to maximize the expected performance during gait trials. Any of the performance metrics used to evaluate the algorithms can be used as the cost function. In this study, we selected the F_1 -score. It is difficult (if possible) to find an analytic form for the algorithms' performance. Therefore, we propose a meta-heuristic method to compute the best set of parameters.

We use a hybrid between two well-known meta-heuristics approaches: Simulated Annealing (SA) and Genetic Algorithm (GA). Both SA and GA are general purpose optimization processes. SA resembles the physical process of heating and slowly cooling a physical solid to minimize the system's energy [35]. At each iteration, the algorithm generates a new point by randomly modifying the current one. Acceptance of

the new points depends on a probability distribution that is a function of the current temperature. An annealing schedule decreases the temperature as the number of iterations increases. GA resembles the natural selection process by iteratively modifying a population of individual solutions. At each iteration (generation), GA uses three operators to create the next generation from the current one: selection, crossover, and mutation [36]. This "evolutionary" process converges the population toward optimal solutions.

For optimization of the model parameters, we use a hybrid SA/GA algorithm as proposed in [37]. It has the same structure of a Genetic Algorithm with the difference that acceptance of the new individuals depends on a probability distribution defined by the current temperature as in SA. This improves the efficiency of the crossover and mutation operators of the GA. Pseudo-code for proposed SA/GA algorithm is shown below.

Algorithm 3 Hybrid SA-GA Algorithm

Input: $ng, ni, T_1, f_{min}, n_{max}$

```

1:  $g = 1$ 
2: Randomly generate an initial population  $S_1$  with  $ni$  sets of parameters.
3: Compute  $Z_1$ : the  $F_1$ -score for each individual.
4: while  $f \geq f_{min}$  or  $n > ng$  do
5:   for  $ind = 1$  to  $ni$  do
6:     Select two individuals  $s_a$  and  $s_b$  from the current population  $S_n$ , using roulette wheel selection.
7:     (CROSSOVER) Recombine the vector entries for  $s_a$  and  $s_b$  to generate a new solution  $S_{n+1}\{ind\}$ .
8:     (MUTATION) Make a random change to  $S_{n+1}\{ind\}$ :  $S_{n+1} = S_{n+1} + \delta S$ .
9:     Compute  $Z_{n+1}\{ind\}$ : the  $F_1$ -score for  $S_{n+1}\{ind\}$ .
10:    if  $Z_{n+1}\{ind\} > Z_n\{ind\}$  then
11:      Compute the acceptance probability,  $P_m$ , by:  $P_m = \exp\{-(Z_{n+1}\{ind\} - Z_n\{ind\})/(T_{n+1})\}$ 
12:      Generate a random number  $0 \leq r \leq 1$ 
13:      if  $P_m > r$  then
14:        Accept the new solution
15:      else
16:        Reject the new solution:  $S_{n+1}\{ind\} = S_n\{ind\}$ 
17:      end if
18:    end if
19:    Decrease  $T$  according to  $T_n = \frac{T_0}{\ln(n)}$ .
20:     $n = n + 1$ 
21:  end for
22:   $g = g + 1$ 
23: end while
Output:  $S_n$ 

```

IV. EXPERIMENTAL SETUP AND PROTOCOL

A. Devices and Experimental Setup

In our previous study [31], we built a pair of wearable devices for gait analysis. We designed them to be worn outside the shoes that subjects use for daily activities. Comprising a

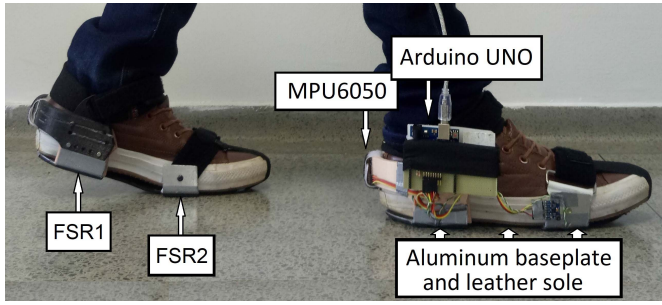


Fig. 2. Wearable devices for gait analysis and therapy.

3D-printed structure with an aluminum baseplate, each one includes a six-axis inertial sensor MPU-6050 (InvenSense, Inc. San Jose, CA, USA). The MPU-6050 was placed on the heel and oriented so that the x-axis was normal to the sagittal plane.

For this study, we are using an improved version of the devices, as shown in Fig. 2. In order to determine the ground truth of the gait event detection, we equipped each one with two force-sensitive resistors (FSRs) to detect contact between foot and ground. We placed the first FSR under the heel and the other one under the second metatarsus. An Arduino UNO board performs data processing in each device and communicates with a PC via USB. A C++ routine stores all gait data in a single file. Sampling rate was set at 50 Hz.

B. Subjects

Subjects were recruited by the Laboratory of Robotic Rehabilitation of the Department of Mechanical Engineering of the University of São Paulo at São Carlos, SP, Brazil. The experimental protocol was approved by the Ethics Committee of the Federal University of São Carlos (#26054813.1.0000.5504). Participants provided informed consent before the session and included: four healthy male subjects without any history of gait abnormalities (with ages ranging between 21 and 33 years old, heights between 1.65 and 1.78 m, and weights ranged between 68 and 83 kg); and, six subjects with idiopathic Parkinson disease in their early clinical stage, four male, two female, aged between 42 and 69 years (58.6 ± 11.2), average height (167 ± 9 cm) and weight (74.0 ± 19.9 kg). Four of them with right dominance. All of them were community ambulators (walking speed 1.15 ± 0.21 m/s during 10 m test). None of the subjects used cane or orthoses to walk. The scores on the modified Hoehn-Yahr scale were between 1 and 1.5 indicating early stage with unilateral and axial involvement. PD subjects who were on medication were tested approximately one hour after medication intake, and were tested in their peak ON state.

C. Experimental Protocol

Subjects wore the wearable device and performed a series of walking trials on treadmill and over ground. Regarding overground trials, at the beginning of each trial, subjects were positioned just behind an initial line marking at the hallway and asked to walk over ground along 15 m roundtrip in self-selected walking speeds (SSWS). PD subjects were able to walk continuously, without freezing-of-gait episodes. After

TABLE I
EXPERIMENTAL PROTOCOL

Subject	Surface	M.S. (m/s)	Trials	Strides	mH&Y
Healthy-1	TM	0.72-1.06	6	278	-
Healthy-2	TM	0.78-1.11	6	283	-
Healthy-3	TM	0.67-0.89	6	251	-
Healthy-4	TM	0.67-1.00	6	313	-
PD-1	OG	~ 1.47	9	77	1
PD-2	OG	~ 1.08	12	159	1
PD-3	OG	~ 0.99	9	103	1.5
PD-4	OG	~ 0.93	9	123	1.5
PD-5	OG	~ 1.13	6	64	1
PD-6	OG	~ 1.13	9	99	1

OG: overground, TM: treadmill, M.S.: Mean speed, mH&Y: Modified H&Y scale.

each round-trip, subjects with PD rested in seated position during one minute, with the option of more time if requested by the participants. Regarding treadmill trials, subjects walked on a treadmill for 60 s, using the first 10 s to reach full speed, 40 s at nominal self-selected speed, 10 s to slow down and stop. Experimental protocol is summarized in Table I.

D. Performance Evaluation and Data Analysis

1) *Data Acquisition and Processing*: All data processing was carried out using MATLAB (The MathWorks, Inc; Natick, MA, USA). To assess the feasibility of real-time identification, we implemented the causal algorithms by passing the data through the algorithms point-by-point, so that only past samples were considered to determine the occurrence of a given event. None of the algorithms requires high computational times or memory usage, so that a causal implementation fairly approximates a real-time setup.

2) *Metrics for Algorithm Evaluation*: We evaluated the accuracy of the event detection algorithms with respect to a sequence of gait phases labeled based on FSR signals. For assessment of the algorithm's accuracy, we computed the F_1 score, Precision (P) and Recall (R) as given by:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2 \cdot PR}{P + R} \quad (8)$$

where F_1 , P and R $\in (0, 1)$, TP are true positives, FP are false positives, and FN are false negatives. We also analyzed the timing agreement between the algorithms and the reference system. Histogram plots were used to determine the time differences between the detection instants. We report the mean values of the differences (MD) with their standard deviations (SD), 95% confidence intervals (CI), and absolute mean difference ($AMD \pm 1$ SD).

3) *Training Procedure*: Training was performed using a jack-knife approach leaving one out for cross validation (LOOCV). In LOOCV, the training set is formed by all the trials of the participants of the group except the trial which will be used for testing the learned mode. This is repeated such that each trial is used once as the testing set. At the end, the F_1 -score of all the trials is averaged.

For Algorithm I, we obtain optimal parameters by:

- Initializing the parameters randomly.

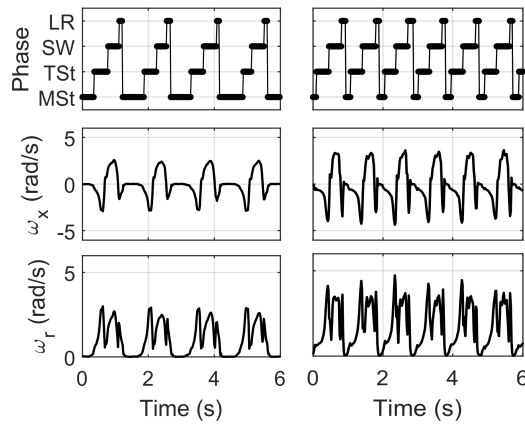


Fig. 3. Representative gait signals for (left) a healthy and (right) a PD subject. Graphs show (top) gait phases labeled using FSR signals, (middle) sagittal angular velocity; (bottom) norm of the angular velocity.

- Optimizing the set of parameters using GA/SA.

For Algorithm II, we obtain optimal parameters by:

- Using feature analysis, to select the features that maximize the mutual information with the label sequence, we ranked available features based on their Fisher scores [38]. The sagittal angular velocity, $\dot{\theta}_x$, and the resultant angular velocity, $\dot{\theta}_r$, were selected since they obtained the highest scores, averaging 0.7 and 1.1, respectively.
- Computing parameters for each classifier with SVM.
- Optimizing the set of parameters using GA/SA.

V. RESULTS

A. Typical Waveforms

Fig. 3 shows representative patterns of the foot angular velocity for the two groups. Note that it is possible to identify similar stride patterns for the same subject (low intra-subject variability of the gait pattern), but they differ substantially between participants (high inter-subject variability). These aspects are relevant to the identification of gait phases, especially because we did not observe direct correspondence to the gait phases with specific values of the inertial signals.

B. Results for Algorithm II: Univariate Classifier

Using SVM, we computed sets of parameters for the univariate linear classifiers that use $\dot{\theta}_x$ and $\dot{\theta}_r$ as input features. Then, starting from the parameter sets computed using SVM, we optimized them using SA/GA algorithm. The meta-heuristic procedure was able to improve the linear classifier performance for all cases, converging to local F_1 maxima in less than ten seconds of CPU time for both groups.

Table II presents the F_1 -scores obtained for the three following classifiers: the rule-based Algorithm I, the univariate linear classifier with SVM-learned parameters, and univariate linear classifier optimized using SA/GA. All optimized classifiers obtained higher values for the healthy group (0.982 ± 0.017) than for the PD group (0.931 ± 0.057). The Algorithm I outperformed the other classifiers (0.988 for the healthy group and

TABLE II
 F_1 SCORE OBTAINED FOR EACH ALGORITHM

	Healthy (n=2320)	PD (n=1322)
Algorithm I. Rule-based		
(with SA/GA)	0.988	0.974
Algorithm II. Linear Classifiers (Univariate)		
<i>Parameters learned with SVM</i>		
using $\dot{\theta}_x$	0.572	0.623
using $\dot{\theta}_r$	0.887	0.852
<i>Parameters optimized with SA/GA</i>		
using $\dot{\theta}_x$	0.991	0.925
using $\dot{\theta}_r$	0.968	0.894
Algorithm II. Linear Classifiers (Multivariate)		
<i>Parameters learned with SVM</i>		
(a) Using $\dot{\theta}_x$ and $\dot{\theta}_r$	0.601	0.795
(b) Using $\dot{\theta}_x$, $\dot{\theta}_y$, and $\dot{\theta}_z$	0.585	0.635
(c) Using $\dot{\theta}_x$, $\dot{\theta}_y$, $\dot{\theta}_z$ and $\dot{\theta}_r$	0.952	0.825
(d) Using $\dot{\theta}_x$, $\dot{\theta}_y$, $\dot{\theta}_z$ and a_r	0.944	0.830
<i>Parameters optimized with SA/GA</i>		
(a) Using $\dot{\theta}_x$ and $\dot{\theta}_r$	0.987	0.953
(b) Using $\dot{\theta}_x$, $\dot{\theta}_y$, and $\dot{\theta}_z$	0.967	0.879
(c) Using $\dot{\theta}_x$, $\dot{\theta}_y$, $\dot{\theta}_z$ and $\dot{\theta}_r$	0.988	0.958
(d) Using $\dot{\theta}_x$, $\dot{\theta}_y$, $\dot{\theta}_z$ and a_r	0.984	0.894

0.974 for the PD group), followed closely by the univariate linear classifier that uses $\dot{\theta}_x$ (0.991 and 0.925 for healthy and PD groups, respectively).

C. Results for Algorithm II: Multivariate Classifier

We evaluated four multivariate classifiers, differing in the features used for event detection:

- Using $\dot{\theta}_x$ and $\dot{\theta}_r$.
- Using $\dot{\theta}_x$, $\dot{\theta}_y$, and $\dot{\theta}_z$.
- Using $\dot{\theta}_x$, $\dot{\theta}_y$, $\dot{\theta}_z$ and $\dot{\theta}_r$.
- Using $\dot{\theta}_x$, $\dot{\theta}_y$, $\dot{\theta}_z$ and a_r , where $a_r = \sqrt{a_x^2 + a_y^2 + a_z^2}$

1) *Performance Results for the Multivariate SVM-Learned Classifiers:* Table II presents the average F_1 -scores obtained by each classifier for each subject. Consider first the classifiers using parameters learned using SVM. Classifiers (c) and (d) obtained better results than the other classifiers. They obtained high performances for the healthy group (0.952 and 0.944). However, they presented an inferior performance for PD group (0.825 and 0.830). Classifiers (a) and (b) presented a very poor performance for both groups (< 0.8). We found that the use of more IMU signals improved the event detection in comparison with the sole use of univariate classifiers.

2) *Performance Results for the Multivariate SA/GA-Optimized Classifiers:* Table II also summarizes the performance of the classifiers optimized using SA/GA. Parameters obtained by this optimization resulted in higher performances of the classifiers. All classifiers reached high values of performance for both groups, similar to the obtained with the Algorithm I and with the univariate classifiers after being optimized. Maximum

TABLE III

PERFORMANCE OBTAINED FOR EACH ALGORITHM DURING TRIALS

	Healthy group (n = 2255)			PD group (n = 1075)		
	P	R	F_1	P	R	F_1
Algorithm I. Heuristic threshold-based method						
HO	0.98	0.98	0.98	0.96	0.98	0.97
TO	0.99	0.99	0.99	0.97	0.99	0.98
HS	0.99	0.99	0.99	0.97	0.99	0.98
TS	0.98	0.98	0.98	0.96	0.98	0.97
	0.988			0.974		
Algorithm II. Linear classifier using $\hat{\theta}_x$ and $\hat{\theta}_r$						
HO	0.98	0.97	0.98	0.99	0.96	0.98
TO	0.99	0.99	0.99	0.92	0.89	0.90
HS	0.99	0.99	0.99	0.99	0.96	0.98
TS	0.99	0.98	0.98	0.97	0.94	0.95
	0.987			0.953		

 F_1 : F_1 -scores; P: Precision; R: Recall.

values were obtained with classifiers (a) and (c) (healthy: 0.988, and PD: 0.956).

D. Comparison of Performance Between Algorithms

Table III details the accuracy in terms of F_1 -score, precision and recall obtained for the algorithms for the four events for the two groups. We present results for the two algorithms: the Algorithm I using threshold-base rules for event detection; and the Algorithm II employing a multivariate linear classifier over $\hat{\theta}_x$ and $\hat{\theta}_r$ signals to detect gait events. Parameters used for the Algorithm I were ascertained using SA/GA, for Algorithm II were learned using SVM and SA/GA. We used LOOCV to train the parameters used in each trial.

For healthy and PD participants gait trials, both algorithms obtained a mean F_1 score > 0.95 . The algorithms showed a high precision (0.95 ± 0.2) indicating that the algorithms are able to avoid false positives. The recall was also high (0.95 ± 0.2) indicating that the algorithms are able to detect most of the gait events. Best performances were obtained for HO and HS events (0.96 ± 0.05 and 0.98 ± 0.03 for all algorithms, respectively). On the other hand, the TO and TS events presented lower accuracy for subjects with PD (0.92 ± 0.10 and 0.94 ± 0.06 , respectively).

Fig. 4 shows the probability histograms of the timing agreement between the proposed algorithms and the FSR-based reference. The positive values represent delays in the algorithm in comparison with the reference. Table IV details the timing detection errors for the two algorithms. In general, detection of HS presented a lower timing error than the other events and detection of TS presented a higher error.

Regarding Algorithm I, average AMD across the four events was 71 ± 62 msec for healthy group and 57 ± 61 msec for PD group. The algorithm tends to have a greater error for TS detection with average AMD values across subjects of 84 ± 37 msec. However the algorithm obtained its lowest MD for this event (-6 ± 124 msec for the healthy subjects, and 8 ± 105 msec for subjects with PD). The algorithm presents a

TABLE IV

TIMING AGREEMENT BETWEEN ALGORITHMS AND REFERENCE

	Healthy group (n = 2255)			PD group (n = 1075)		
	MD	CI	AMD	MD	CI	AMD
Algorithm I. Heuristic threshold-based method						
HO:	35 ± 82	[31, 38]	65 ± 61	25 ± 97	[20, 30]	73 ± 69
TO:	53 ± 99	[49, 57]	89 ± 69	8 ± 66	[4, 11]	43 ± 50
HS:	-14 ± 45	[-16, -12]	32 ± 35	-4 ± 66	[-8, -1]	39 ± 54
TS:	-6 ± 124	[-11, -1]	97 ± 76	8 ± 105	[2, 14]	71 ± 78
Algorithm II. Linear classifier using $\hat{\theta}_x$ and $\hat{\theta}_r$						
HO:	61 ± 66	[58, 63]	74 ± 50	0 ± 103	[-6, 5]	83 ± 62
TO:	48 ± 96	[44, 52]	85 ± 66	33 ± 88	[28, 38]	60 ± 73
HS:	27 ± 41	[26, 29]	40 ± 30	48 ± 68	[45, 52]	59 ± 59
TS:	81 ± 102	[77, 85]	111 ± 68	28 ± 110	[22, 35]	92 ± 67

MD: Mean difference; CI: 95% confidence intervals; AMD: absolute mean difference. Values in ms.

lower error in the timing agreement for the HS detection with average AMD values across subjects of 36 ± 37 msec.

For Algorithm II, average AMD values across events were 78 ± 55 msec for the healthy group and 74 ± 62 msec for the PD group. Larger AMD values across subjects were observed for TS event (102 ± 68 msec). Lower timing errors were obtained for HS event (49 ± 52 msec).

VI. DISCUSSION

The aim of this study was to develop and assess two algorithms for real-time gait event detection on participants with gait impairments. To assess the performance of the newly proposed algorithms in detecting gait events during walking, the sequence of gait phases based on FSR signals was used as a reference method. Experimental results demonstrate the reliability of the proposed algorithms on detecting gait events based on inertial data measured at the subject's foot. As detailed in the previous section, the algorithms presented good detection results for subjects with PD with all $F_1 > 0.90$.

A. Regarding the Proposed Methods: Algorithms, Parameter Selection and Adaptation

Use of Linear Classifiers is a new method that we propose for gait event detection; it can be understood as a generalization of the threshold based approach. It is important to highlight that this approach allows the use of any feature without previous knowledge of the gait patterns. Despite its simplicity, linear classifiers obtained same performance as the other well-established approaches, such as threshold-based rules, with a level of robustness and accuracy that allow its implementation in real-time applications.

Speed variations during gait require modification of the optimal thresholds in order to maintain high performance and timing [39]. Machine learning approaches have shown superior performance to determine optimal parameters and rules than human experts. However, we must study further on how often and when the system must be retrained in order to keep tracking changing conditions during therapy [40]. The proposed methodology is part of our ongoing efforts to develop a participant specific rehabilitation approach.

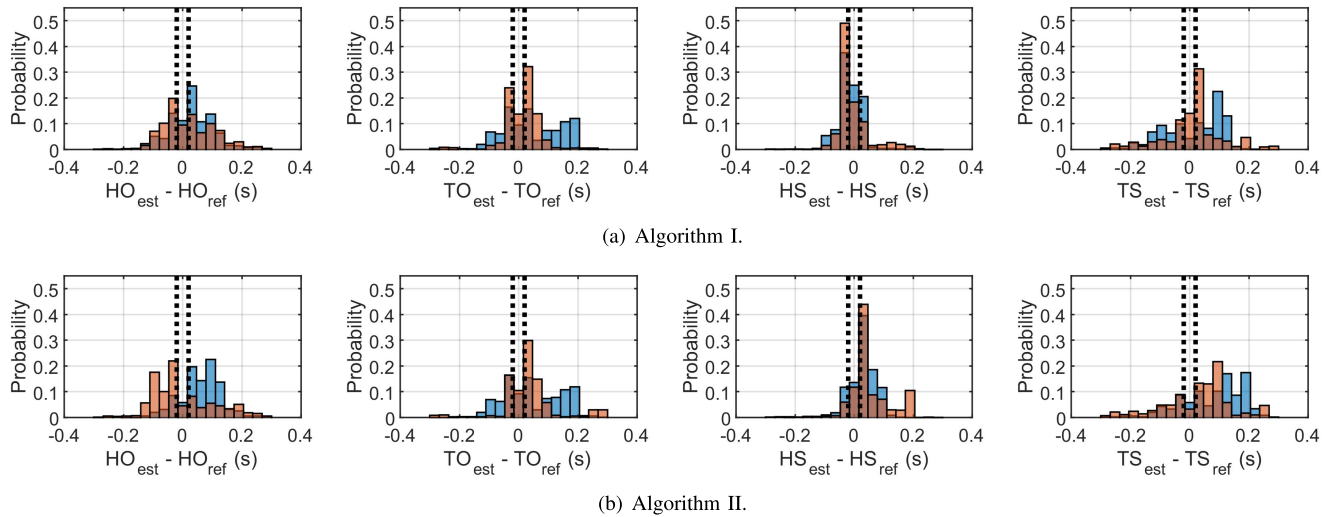


Fig. 4. Normalized frequency distribution of the time differences between the event detection of the two proposed algorithms and the reference system for (left) HO, (left-center) TO, (right-center) HS, and (right) TS gait events. Bar colors indicate the subject: healthy (blue) and Parkinsonian (red). Positive times correspond to delays in the proposed algorithm. Vertical dotted lines correspond to the period given by the sampling frequency of the acquisition system, $f_c = 50$ Hz.

B. Comparison With Other Methods

The results presented in this paper can be compared to the results obtained with previous methods evaluated with impaired subjects. Similar to previous approaches, our algorithms tend to obtain lower performances for impaired subjects than for healthy ones. In [40], Rueterbories *et al.*, reported for 10 healthy subjects an average detection rate of 102.5% (more than 100% indicates multiple detection of the same event), with median recall values of 0.96 for HS/IC, and 1 for the other three events. For 10 hemiparetic subjects an average detection rate of 126.8%, with median recall values of 0.81 for HS/IC, and 1 for the other three events was determined. From their report, it can be observed that performance (measured in F1 score) drops from a mean F1 score across events for healthy subjects of 0.98, to 0.83 for hemiparetic subjects. Despite having high recall (near 1), their algorithm was subject to a high incidence of false positives (precision ≈ 0.7). They also reported increases in the AMD values from healthy to hemiparetic subjects: from 43.8 to 65.3 msec for HS detection, from 43.4 to 104.1 msec for TS, from 59.5 to 73.6 ms for HO, and from 53.9 to 70.7 msec for TO.

In [11], Mariani *et al.*, identify a set of features and thresholds for 42 subjects, including healthy subjects and patients, before and after surgical treatments for ankle osteoarthritis. For HS they reported AMD of 8 msec and MD of -1 msec when looking for minimum peaks in $\ddot{a} = d(\dot{\theta}_x)/dt$. For TS, the best results were obtained when detecting $\dot{\theta}_x > -2$ rad/s, showing MD of -4 msec and AMD of 31 msec, respectively. For HO, they obtained MD and AMD values of 4 and 46 ms respectively when looking at low $\ddot{a}$, and a MD value of 76 msec when $\dot{\theta}_x$ was below a given threshold (-2 rad/s). For TO, they obtained MD and AMD values of -3 and 11 msec when detecting the event at the maximum peak of $\ddot{a}$. In [10], Abaid *et al.*, reported mean specificity and sensitivity greater than or equal to 0.95 in the detection of the four gait events during walking trials of children with hemiplegia. They

used a stochastic model of the gait called Hidden Markov Model.

C. Perspectives on Automatic Detection of Gait Phases

Gait is a regular and quasi-cyclic process that can be divided into several phases, and each one of these phases has an specific function in the walking biomechanics. Automatic detection of these phases, and of the events that define them, is a major step for diverse areas of the human biomechanics field. For example, it allows the segmentation in strides of the signals produced during human walking, enabling the parameterization and spatiotemporal normalization of these signals. That is useful for exploratory data analysis and for summary statistics because it provides information and features (e.g. number of strides, cadence, stride duration) generally used for diagnosis and evaluation of gait abnormalities.

Robot-aided therapies are increasingly used in gait rehabilitation and have shown positive impacts on patient recovery. Tailoring therapy assistance to the particular patient's needs is an important feature of training (Assistance-As-Needed paradigm). In [41], Pérez-Ibarra *et al.*, proposed an adaptive impedance controller that modified the amount of robotic assistance/resistance according to the estimation of the human contribution during open-ended movements. During walking, in addition to the appropriate amount of force, an assist-as-needed therapy also requires to determine the right moment to apply or remove that force. In this context, the appropriate real-time identification of gait events and phases becomes a relevant issue.

D. Limitations and Recommendations for Further Research

One needs to take our results with the appropriate caveats, in particular, because the number of subjects is small. In the case of the PD subjects, the absence of comparison of

the algorithms' performance between ON and OFF states must be considered before any interpretation of the results. Nevertheless, this study demonstrates the feasibility of the proposed scheme to learn sets of parameters tuned to different groups of subjects with similar pathology. Our results support larger clinical studies involving other conditions to determine whether the proposed strategy can be generalized.

Another potential drawback of this study is the reference system. The use of FSR signals for detection of gait phases presents some limitations that may contribute to the observed errors in performance and timing.

VII. CONCLUSION

In this paper, we proposed a set of algorithms for online identification of gait phases and events. The first algorithm detects events through a heuristic threshold-based method. Then, we proposed the use of linear classifiers as a generalization of the threshold-based approach. It is important to highlight that this approach allows the use of any feature without previous knowledge of the signal patterns during walking. Both algorithms proposed are parametric, i.e., they use a set of numerical values to decide when a transition occurs. To compute the optimal set of parameters for each subject, we proposed a parameter learning procedure using Support Vector Machines, and a parameter optimization procedure using a meta-heuristic method. In addition, we developed an improved version of our IMU-based wearable device that allows measurement and storage of gait data. We tested the proposed algorithms and device in healthy and PD participants while walking over ground and on a treadmill. Experimental results showed the reliability of the proposed algorithms on detecting gait events based on inertial data measured at the subject's foot. Linear Classifiers is a new method for gait event detection. Despite its simplicity, it obtained the same performance as the threshold-based algorithm, with robustness and accuracy that allow its implementation in real-time applications.

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